

Machine Learning (2025)

Homework 2: Classification

Deadline: 2025/4/22



In this homework, you are asked to use linear models to classify handwritten digits from the MNIST dataset in two different ways:

1. Binary Classification: Predict whether a digit is Odd or Even.
2. Multiclass Classification: Predict the exact digit (0-9).

Training Data: 10,000 images (28×28 grayscale), with 1,000 samples per digit.

Testing Data: 2,000 images.

```
hw2/
├── train/
│   ├── 0/
│   │   ├── 0.jpg
│   │   ├── 1.jpg
│   │   └── ...
│   ├── 1/
│   │   ├── 0.jpg
│   │   ├── 1.jpg
│   │   └── ...
│   └── ...
└── test/
    ├── 0.jpg
    ├── 1.jpg
    └── ...
```

1. Binary Classification (30%)

Train a binary logistic regression model to predict whether a digit is odd (1, 3, 5, 7, 9) or even (0, 2, 4, 6, 8).

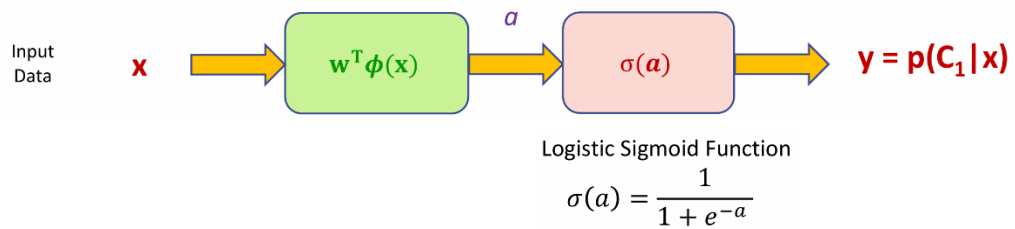
Output labels: 0 (even), 1 (odd).

Requirement: accuracy $\geq 85\%$.

[Input] train/*/*.jpg, test/*/*.jpg

[Output] result_1.csv

Linear Model



$$E(\mathbf{w}) = -\ln p(\mathbf{t}|\mathbf{X}, \mathbf{w}) = -\sum_{n=1}^N \{t_n \ln y(\mathbf{x}_n, \mathbf{w}) + (1 - t_n) \ln(1 - y(\mathbf{x}_n, \mathbf{w}))\}$$

$$\mathbf{W}^{(new)} = \mathbf{W}^{(old)} - \eta \nabla E(\mathbf{W}^{(old)}) \quad \text{Gradient Descent}$$

$$\mathbf{W}^{(new)} = \mathbf{W}^{(old)} - \mathbf{H}^{-1} \nabla E(\mathbf{W}^{(old)}) \quad \text{Newton-Raphson}$$

$$\text{where } \mathbf{H} = \nabla \nabla E(\mathbf{W})$$

2. Multi-Class Classification (30%)

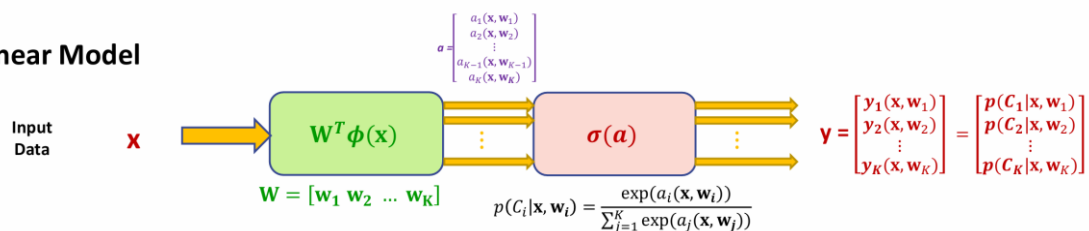
Train a multi-class classification model to identify the digit (0-9).

Requirement: accuracy $\geq 85\%$.

[Input] train/*/*.jpg, test/*/*.jpg

[Output] result_2.csv

Linear Model



$$E(\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_K) = -\ln p(\mathbf{T}|\mathbf{X}, \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_K)$$

$$= -\sum_{n=1}^N \sum_{k=1}^K t_{nk} \ln y_k(\mathbf{x}_n; \mathbf{w}_k)$$

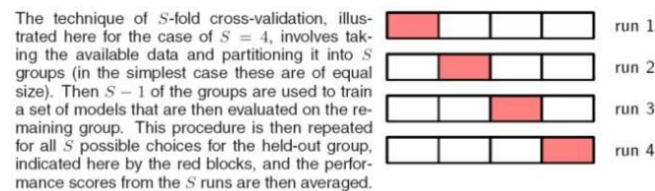
$$\mathbf{W}^{(new)} = \mathbf{W}^{(old)} - \eta \nabla E(\mathbf{W}^{(old)}) \quad \text{Gradient Descent}$$

$$\mathbf{W}^{(new)} = \mathbf{W}^{(old)} - \mathbf{H}^{-1} \nabla E(\mathbf{W}^{(old)}) \quad \text{Newton-Raphson}$$

$$\text{where } \mathbf{H} = \nabla \nabla E(\mathbf{W})$$

S-Fold Cross-Validation:

Since you will only receive the training dataset, you can use this cross-validation technique to evaluate your model's accuracy.



Provided code:

problem_1.py / problem_2.py

```

1  import numpy as np
2  from PIL import Image
3  import os
4
5  Codeium: Refactor | Explain | Generate Docstring | X
6  def read_train_data():
7      data_X = np.zeros((0, 28*28))
8      data_T = np.zeros((0, 1), dtype=int)
9      for i in range(10):
10         for j in range(1000):
11             img = np.array(Image.open(f"train/{i}/{j}.jpg"))
12             data_X = np.vstack([data_X, img.reshape((1, 28*28))])
13             data_T = np.vstack([data_T, i%2])
14         return data_X, data_T
15
16  Codeium: Refactor | Explain | Generate Docstring | X
17  def read_test_data():
18      data_X = np.zeros((0, 28*28))
19      data_T = np.zeros((0, 1), dtype=int)
20      for i in range(2000):
21         img = np.array(Image.open(f"test/{i}.jpg"))
22         data_X = np.vstack([data_X, img.reshape((1, 28*28))])
23         data_T = np.vstack([data_T, i%2])
24     return data_X
25
26  Codeium: Refactor | Explain | Generate Docstring | X
27  def save_result(y_pred):
28     if not os.path.exists('./outputs'):
29         os.makedirs('./outputs')
30     with open('./outputs/result_1.csv', 'w') as f:
31         for i in range(len(y_pred)):
32             f.write(f"{y_pred[i, 0]}\n")

```

Homework Rules and Grading Policy

Homework will be graded by:

1. Program Correctness (60%)

For each problem, the testing **accuracy** of your prediction must be **greater than 85%**. (TA will provide an additional testing dataset to help you verify that your output format is correct.)

2. Report (40%)

- Binary logistic regression
- Multi-class classification
- Result
- Discussion

Remind:

- **Program:** Please use **Python** and only the **NumPy** package for regression models.
- **Report:** Use font size **12**, with a maximum of **7** pages. Explaining the key aspects of your approaches and visualizing your experimental results will earn a higher score on the report.
- **Deadline:** Late submissions within **3** days receive 70% of the score. Submissions beyond **3** days are NOT accepted.
- **Test Failure:** If you do not pass the correctness check, half of the score will be deducted, but there is a chance for remediation.
- **Plagiarism:** If two codes show high similarity, both will receive zero marks after confirmation.

Upload:

[Web] E3

[File Name] hw2_{student id}.zip

(e.g., hw2_123456789.zip)

```
hw2_{student_id}/
├── train/
├── (test/)
├── outputs/
│   ├── (result_1.csv)
│   └── (result_2.csv)
├── report.pdf
├── problem_1.py
├── problem_2.py
└── README.txt
```